

1) What is a Parameter in Tableau and How Does it Differ from a Filter?

A **Parameter** in Tableau is a user-defined variable that allows users to dynamically change values in calculations, reference lines, filters, and visualizations. It accepts a single value at a time and can be controlled by the user through a parameter control.

A **Filter**, on the other hand, is used to limit the data displayed in a visualization by including or excluding specific records.

Difference Between Parameter and Filter

Parameter	Filter
User-defined variable	Data restriction tool
Accepts a single value at a time	Can select one or multiple values
Can be used in calculations and logic	Only filters data displayed
Does not directly filter data unless linked to a calculation	Directly filters the data
Remains available even if the value doesn't exist in the data	Shows values present in the data source

Example:

- Parameter: User selects a year (2023 or 2024) to compare metrics dynamically.
- Filter: User filters data to show only records from 2024.

2) List Three Use Cases Where Parameters Are Preferred Over Filters

1. Dynamic Measure Selection

Users can switch between different measures such as:

- Sales
- Profit
- Quantity

using a single parameter instead of creating multiple charts.

2. What-If Analysis

Parameters allow users to enter hypothetical values and analyze outcomes.

Example:

A hospital administrator changes the expected bed occupancy rate from 80% to 90% and observes the impact on capacity planning.

3. Dynamic Reference Lines or Targets

Parameters can control benchmark values in charts.

Example:

Users can set a target patient satisfaction score and compare actual performance against the selected target.

3) What Are the Four Data Types Supported in Tableau Parameters?

Tableau supports the following four parameter data types:

1. Integer Stores whole numbers. Example: 100 250 500	2. Float Stores decimal numbers. Example: 4.5 75.25 99.99	3. String Stores text values. Example: Sales Profit Revenue	4. Date Stores date values. Example: 01-Jan-2025 15-Feb-2025 31-Dec-2025	Summary <table><tr><th>Data Type</th><th>Example</th></tr><tr><td>Integer</td><td>100</td></tr><tr><td>Float</td><td>99.5</td></tr><tr><td>String</td><td>Sales</td></tr><tr><td>Date</td><td>01-Jan-2025</td></tr></table>	Data Type	Example	Integer	100	Float	99.5	String	Sales	Date	01-Jan-2025
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